

N-Nitroso compounds and cancer incidence: the European Prospective Investigation into Cancer and Nutrition (EPIC)-Norfolk Study.

BACKGROUND: Humans are exposed to preformed N-nitroso compounds (NOCs) and endogenous NOCs. Several NOCs are potential human carcinogens, including N-nitrosodimethylamine (NDMA), but evidence from population studies is inconsistent. **OBJECTIVE:** We examined the relation between dietary NOCs (NDMA), the endogenous NOC index, and dietary nitrite and cancer incidence in the European Prospective Investigation into Cancer and Nutrition (EPIC)-Norfolk, United Kingdom, study. **DESIGN:** This was a prospective study of 23,363 men and women, aged 40-79 y, who were recruited in 1993-1997 and followed up to 2008. The baseline diet was assessed with food-frequency questionnaires. **RESULTS:** There were 3268 incident cancers after a mean follow-up of 11.4 y. Dietary NDMA intake was significantly associated with increased cancer risk in men and women [hazard ratio (HR): 1.14; 95% CI: 1.03, 1.27; P for trend = 0.03] and in men (HR: 1.24; 95% CI: 1.07, 1.44; P for trend = 0.005) when the highest quartile was compared with the lowest quartile in age- and sex-adjusted analyses but not in multivariate analyses (HR: 1.10; 95% CI: 0.97, 1.24; HR for men: 1.18; 95% CI: 1.00, 1.40; P for trend $\geq$ 0.05). When continuously analyzed, NDMA was associated with increased risk of gastrointestinal cancers (HR: 1.13; 95% CI: 1.00, 1.28), specifically of rectal cancer (HR: 1.46; 95% CI: 1.16, 1.84) per 1-SD increase after adjustment for age, sex, body mass index, cigarette smoking status, alcohol intake, energy intake, physical activity, education, and menopausal status (in women). The endogenous NOC index and dietary nitrite were not significantly associated with cancer risk. There was a significant interaction between plasma vitamin C concentrations and dietary NDMA intake on cancer incidence (P for interaction < 0.00001). **CONCLUSIONS:** Dietary NOC (NDMA) was associated with a higher gastrointestinal cancer incidence, specifically of rectal cancer. Plasma vitamin C may modify the relation between NDMA exposure and cancer risk.